

Muhang Tian

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EDUCATION	Courant Institute, New York University Ph.D. in Computer Science	01/2025 – Expected 12/2029
	Duke University B.S. in Computer Science, double minor in Mathematics and Economics <i>Graduation with Highest Distinction, Alex Vasilos Memorial Award</i>	08/2020 – 05/2024

PUBLICATIONS	[1] Reward-rate Policy Gradient for Efficient Machine Learning Engineering Agents	2026
	<u>Muhang Tian</u> , Sherry Yang <i>Preprint</i>	
	[2] Multi-objective Reinforcement Learning with Nonlinear Preferences: Provable Approximation for Maximizing Expected Scalarized Return	2025
	Nianli Peng*, <u>Muhang Tian</u> *, Brandon Fain <i>Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)</i>	
	[3] Reliable Generation of Privacy-preserving Synthetic Electronic Health Record Time Series via Diffusion Models	2024
	<u>Muhang Tian</u> , Bernie Chen, Allan Guo, Shiyi Jiang, Anru R Zhang <i>Journal of the American Medical Informatics Association (JAMIA)</i>	
	[4] Fast and Interpretable Mortality Risk Scores for Critical Care Patients	2024
	Chloe Qinyu Zhu*, <u>Muhang Tian</u> *, Lesia Semenova, Jiachang Liu, Jack Xu, Joseph Scarpa, Cynthia Rudin <i>Journal of the American Medical Informatics Association (JAMIA)</i>	
	[5] Welfare and Fairness in Multi-objective Reinforcement Learning	2023
	Zimeng Fan*, Nianli Peng*, <u>Muhang Tian</u> *, Brandon Fain <i>Proceedings of the International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)</i>	

WORK EXPERIENCE	ByteDance <i>Research Intern</i>	06/2026 – 09/2026
	Explored an agentic framework to automate research experiments for generative recommender systems.	
	Big Ideas Lab <i>Machine Learning Engineer</i>	05/2022 – 03/2023
	Applied EfficientDet for object detection tasks on COVID-19 test results.	
	Afloat <i>Software Engineer</i>	05/2022 – 08/2022
	Maintained company server, performed software testing, and reported weekly directly to co-founders.	

RESEARCH EXPERIENCE	K-space Transformer	03/2025 – 03/2026
	■ Explored using Vision Transformers to learn meaningful representations from k-space, aiming to perform prediction directly from Fourier transforms of natural images. ■ Explored on ImageNet with different training recipes and normalization schemes. Observed that using a linear layer for tokenization is crucial for accurate prediction.	

- Explored generating mixed-type time series data with multinomial and Gaussian diffusions using bidirectional recurrent neural network as the backbone.
- Experimentally demonstrated the proposed approach produces synthetic samples with high data utility, outperforms generative adversarial networks, and improves ML models’ performance.

- Explored optimizing the expectation over a nonlinear objective on the cumulative reward. Proposed an approximate algorithm that optimizes this objective by considering past accumulated rewards.
- Experimentally and theoretically verified that the algorithm converges. Designed another dynamic programming approach to learn a policy with theoretical guarantees.

- Explored finding an interpretable model for classification task through approximately optimizing a L0 regularized logistic regression model with integer, sparsity, and box constraints.
- Designed group sparsity and monotonicity constraints to improve readability and usefulness of the risk score in practical settings.

- Researched Sample Average Uncertainty approach towards multi-arm bandit problems by utilizing approximate sample mean variance as an exploration bonus.
- Implemented Deep Q-Learning, Proximal Policy Optimization (PPO), and Advantage Actor-Critic algorithms from scratch and compared their performances in Atari environments.

HONORS AND AWARDS	Henry M. MacCracken Award: multi-year funding for New York University PhD students.2024 CS+ Summer Research Fellowship: Duke summer research funding (\$10,000).2022 – 2023
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ACADEMIC SERVICES	Reviewer: NeurIPS (2023), ICLR (2024), AAAI (2025), IEEE ICHI (2023). Teaching Assistant: Fundamentals of Machine Learning, Design and Analysis of Algorithms.
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TECHNICAL SKILLS	Languages: Python, Java, JavaScript, C/C++, R, SQL, MATLAB, CSS/HTML. Softwares: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Linux, Slurm, Docker, Git.
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